QA Embedded — Hardware Test Sheet / Test Log

Smart Gate Controller

• Wokwi • Black-box testing

Test Session

Tester: Mariia Alieksieievych		**Date 22/09/2026**	
Build / Firmware		Environment	Wokwi
Project	Smart Gate Controller	Board	Arduino Uno
Test Type	Functional / Regression / Exploratory	Result	

Test Execution Rules

- Test the device as a black box.
- Record observable behavior, not assumptions about the implementation.
- For every FAIL, create a Bug Report.
- Use the same test cases for regression after firmware changes.
- Try both positive and negative scenarios.

Test Cases

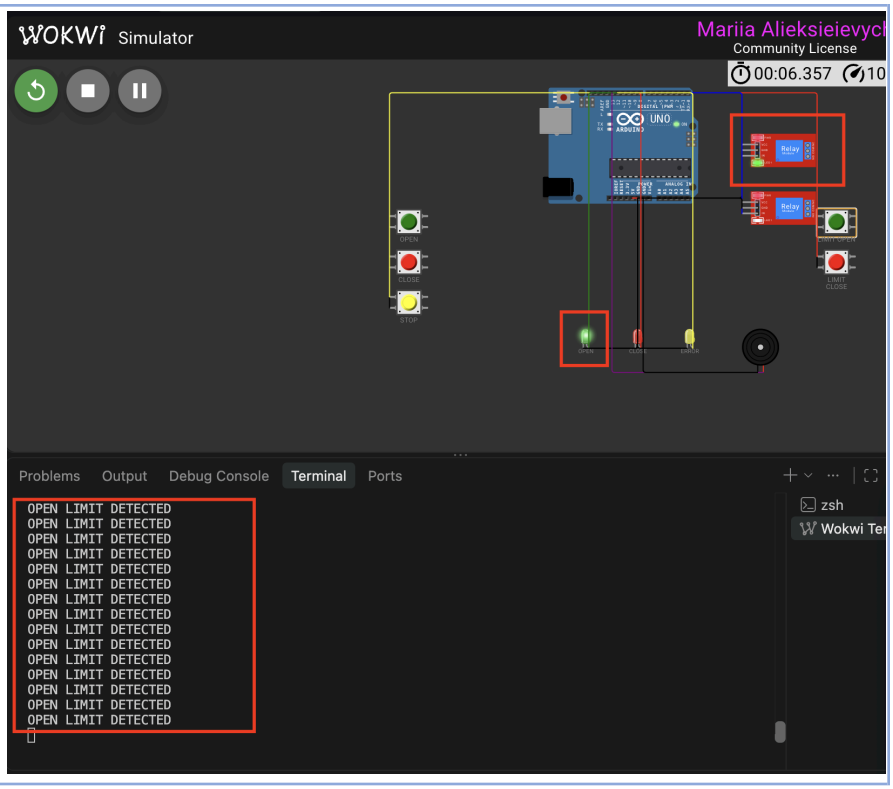
ID	Requirement	Precondition	Action / Steps	Expected Result	Actual Result	P/F	Bug ID
TC-01	R1	After startup	Start simulation	State CLOSED; relays OFF	State CLOSED; relays OFF	PASS	
TC-02	R2	CLOSED	OPEN	OPENING; OPEN relay ON; OPEN LED ON	OPENING; OPEN relay ON; OPEN LED ON	PASS	
TC-03	R3	OPENING	Activate LIMIT_OPEN	OPEN; OPEN relay OFF	Endless OPEN LIMIT DETECTED is displayed; OPEN relay ON; OPEN LED ON	FAIL	BUG-001
TC-04	R4	OPEN	CLOSE	CLOSING; CLOSE relay ON; CLOSE LED ON	BLOCKED by BUG-001	BLOCKED	
TC-05	R5	CLOSING	Activate LIMIT_CLOSE	CLOSED; CLOSE relay OFF	CLOSED; CLOSE relay OFF	PASS	
TC-06	R6	OPENING	STOP	STOPPED; both relays OFF	STOPPED; both relays OFF; **OPEN LED ON**	FAIL	BUG-005
TC-07	R6	CLOSING	STOP	STOPPED; both relays OFF	STOPPED; both relays OFF; **OPEN LED ON**	FAIL	BUG-005
TC-08	R7	OPENING	OPEN again	No unsafe/contradictory state	IGNORED: OPEN is not valid in current state	PASS	

TC-09	R7	CLOSING	CLOSE again	No unsafe/contradictory state	IGNORED: CLOSE is not valid in current state	PASS	
TC-10	R8	CLOSING	OPEN	Safe transition; never both relays ON	Missing safe stop → safety pause. OPEN relays ON	FAIL	BUG-006
TC-11	R8	OPENING	CLOSE	Safe transition; never both relays ON	Missing safe stop → safety pause. CLOSE relays ON	FAIL	BUG-006
TC-12	R9	Any moving state	Observe relay combination	Never OPEN+ CLOSE relay simultaneously	The relays do not activate simultaneously	PASS	
TC-13	R10	OPENING	Wait >10 s without open limit	ERROR; relays OFF; error indication	ERROR is OFF; OPEN relay ON; OPEN LED ON	FAIL	BUG-004
TC-14	R10	CLOSING	Wait >10 s without close limit	ERROR; relays OFF; error indication	ERROR; relays OFF; error indication	PASS	
TC-15	R11	Any valid state	Activate both limits	ERROR; relays OFF; error indication	WARNING: both limit switches active; relays OFF; error indication OFF	FAIL	BUG-003
TC-16	R12	ERROR	OPEN	No movement; relays OFF	No movement; relays OFF	PASS	
TC-17	R12	ERROR	CLOSE	No movement; relays OFF	No movement; relays OFF	PASS	
TC-18	R13	ERROR	RESET	STOPPED; relays OFF	STOPPED; relays OFF; **OPEN LED ON**	FAIL	BUG-005
TC-19	R14	CLOSED	Send invalid command, e.g. X	No movement; diagnostic message	No movement; UNKNOWN COMMAND diagnostic message	PASS	
TC-20	R15	OPENING	Observe LEDs/relay	OPEN LED ON; CLOSE LED OFF; OPEN relay ON	OPEN LED ON; CLOSE LED OFF; OPEN relay ON	PASS	
TC-21	R15	CLOSING	Observe LEDs/relay	CLOSE LED ON; OPEN LED OFF; CLOSE relay ON	CLOSE LED ON; OPEN LED OFF; CLOSE relay ON	PASS	
TC-22	R15	ERROR	Observe outputs	ERROR LED ON; relays OFF	ERROR LED ON; relays OFF	PASS	

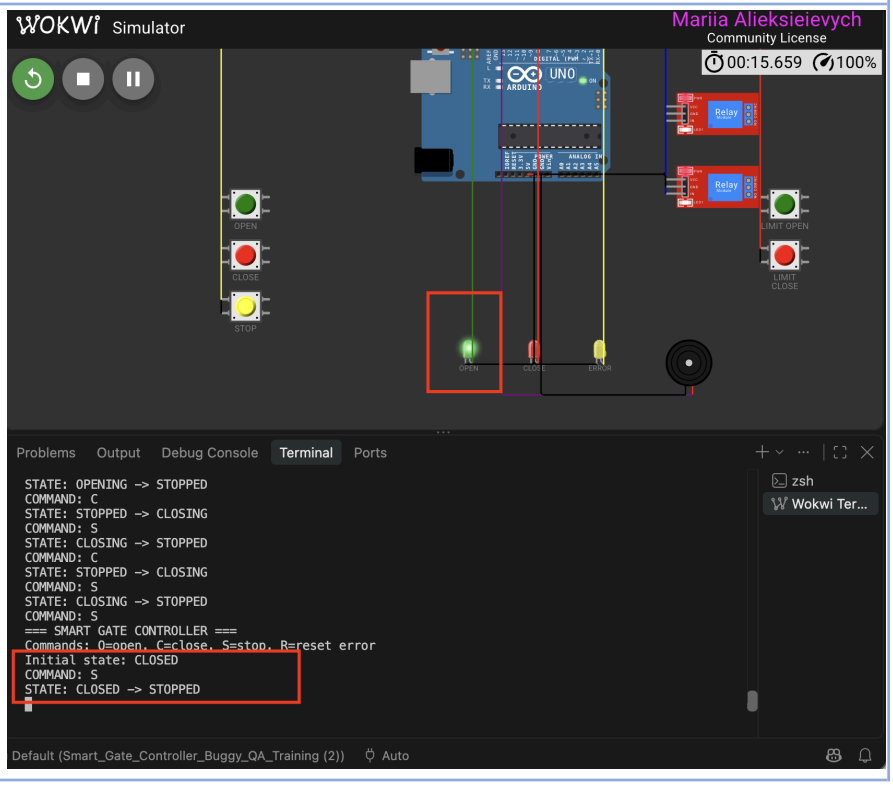
TC-23	R16	CLOSED	Single OPEN press	One OPEN command / one state transition	One OPEN command / one state transition; STATE: CLOSED -> OPENING	PASS	
TC-24	R16	CLOSED	Hold OPEN button	No uncontrolled repeated commands	No movement; No uncontrolled repeated commands	PASS	

Bug report

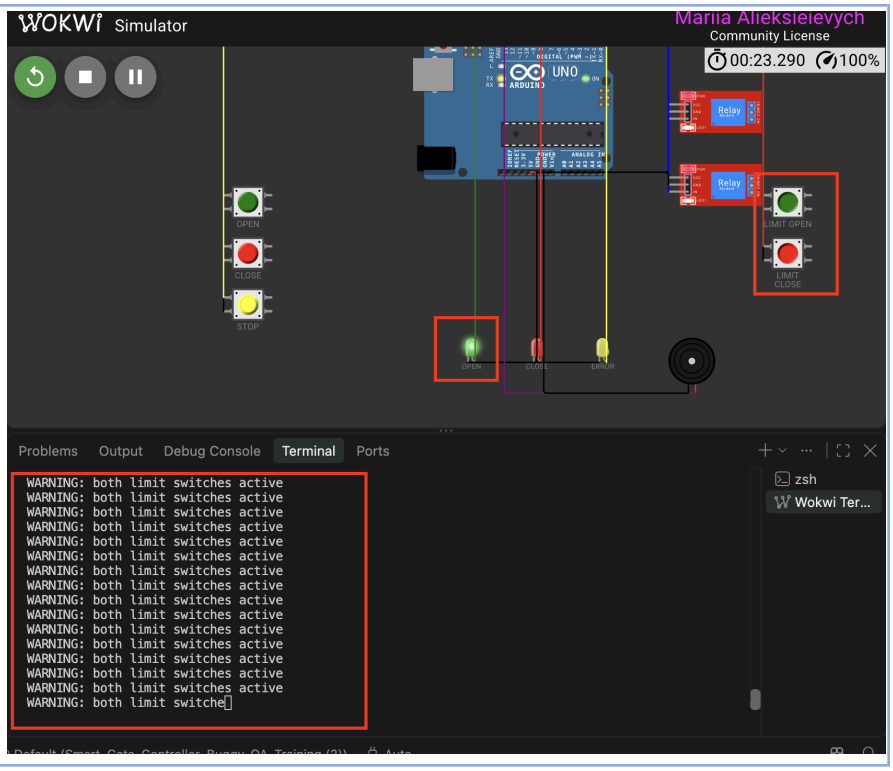
Поле	Значення
Bug ID	BUG-001
Title	OPEN relay remains ON after tapping LIMIT OPEN
Requirement	R6. LIMIT_OPEN під час OPENING вимикає OPEN relay і переводить систему в OPEN.
Precondition	Open the project in VS Code and launch the Wokwi simulator. The system is in the CLOSED state.
Steps	1. Tap OPEN button. 2. Tap LIMIT OPEN
Expected	OPEN relay is OFF; The system is in the OPEN state.
Actual	Endless OPEN LIMIT DETECTED is displayed; OPEN relay is ON; The system is not in the OPEN state.
Severity	Critical
Reproducibility	5/5
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi Simulator, Arduino Uno

Evidence / Notes	

Поле	Значення
Bug ID	BUG-002
Title	The system transitions from CLOSED to STOPPED
Requirement	R5. STOP у будь-якому стані руху негайно вимикає обидва реле та переводить систему у STOPPED. Уточнення від ментора Олега: STOP зупиняє ворота тільки під час руху. У стабільних станах CLOSED та OPEN команда STOP не змінює поточний стан.
Precondition	Open the project in VS Code and launch the Wokwi simulator. The system is in the CLOSED state.
Steps	1. Tap STOP
Expected	The system remains in the CLOSED state; both relays are OFF; all LEDs are OFF.
Actual	The system state is STOPPED; both relays are OFF; LED OPEN is ON
Severity	Medium

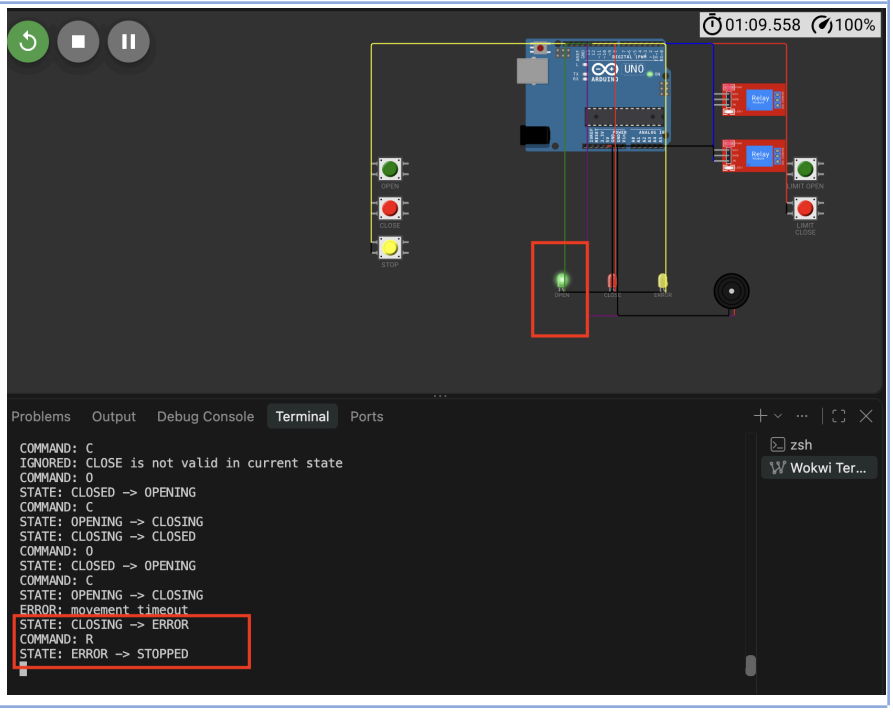
Reproducibility	5/5
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi Simulator, Arduino Uno
Evidence / Notes	

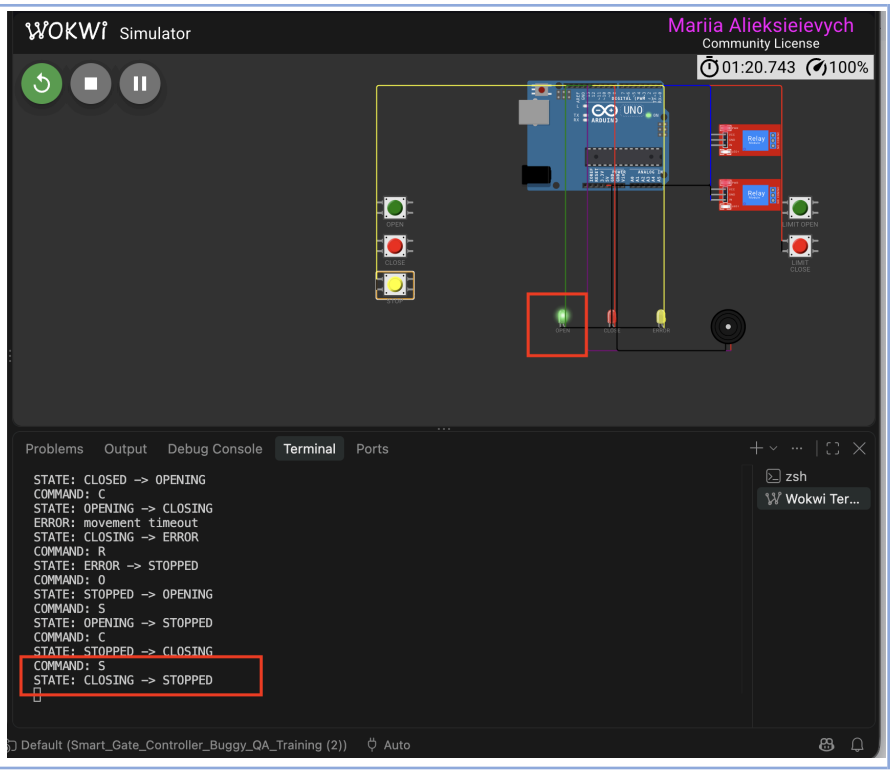
Поле	Значення
Bug ID	BUG-003
Title	ERROR state is not triggered when both limit switches are active
Requirement	R10. Одночасно активні LIMIT_OPEN і LIMIT_CLOSE — ERROR.
Precondition	Open the project in VS Code and launch the Wokwi simulator. The system is in the CLOSED state.
Steps	1. Tap Open 2. Tap LIMIT OPEN and LIMIT CLOSE at the same time.
Expected	The system is in the ERROR state; ERROR LED is ON; Buzzer is active
Actual	The system is not in the ERROR state; ERROR LED is OFF; Buzzer is not active; Endless warning is displayed; OPEN LED is ON
Severity	Major
Reproducibility	5/5
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi Simulator, Arduino Uno

Evidence / Notes	

Поле	Значення
Bug ID	BUG-004
Title	OPENING state lasts more than 10 seconds without ERROR
Requirement	R9. Рух довше 10 секунд без досягнення кінцевика переводить систему в ERROR.
Precondition	Open the project in VS Code and launch the Wokwi simulator. The system is in the CLOSED state.
Steps	1. Tap OPEN button. 2. Wait for more than 10 seconds
Expected	The system state is ERROR; ERROR LED is ON; Buzzer is active
Actual	The system state is OPENING; ERROR LED is OFF; Buzzer is not active; OPEN LED is ON
Severity	Critical
Reproducibility	5/5
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi Simulator, Arduino Uno
Evidence / Notes	Video: https://drive.google.com/file/d/1G2BjAUIofNxH1y9PH0iSNChAqycPJn4X/view?usp=sharing

Поле	Значення

Bug ID	BUG-005
Title	OPEN LED is ON after RESET and STOP
Requirement	R14. OPEN LED відповідає стану OPEN/OPENING; CLOSE LED — CLOSING; ERROR LED — ERROR.
Precondition	Open the project in VS Code and launch the Wokwi simulator. The system is in the CLOSED state.
Steps	RESET: 1. Tap Open button 2. Tap Close button 3. Wait for 10 seconds 4. Tap RESET after ERROR STOP (Case 1): 1. Tap Open button 2. Tap Close button 3. Tap STOP STOP (Case 2): 1. Tap Open button 2. Tap Close button 3. Tap Open button 4. Tap STOP
Expected	All LEDs are OFF
Actual	OPEN LED is ON
Severity	Minor
Reproducibility	5/5
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi Simulator, Arduino Uno
Evidence / Notes	

Поле	Значення
Bug ID	BUG-006
Title	Executing OPEN during CLOSING and CLOSE during OPENING is unsafe
Requirement	R8. Обидва реле ніколи не можуть бути активними одночасно.
Precondition	Open the project in VS Code and launch the Wokwi simulator. The system is in the OPENING/CLOSING state.
Steps	FOR CLOSE during OPENING: 1. Tap OPEN 2. Tap LIMIT OPEN 3. Tap CLOSE For OPEN during CLOSING: 1. Tap OPEN 2. TAP CLOSE 3. TAP LIMIT CLOSE 4. Tap OPEN
Expected	No contradictory relay state; safe transition.
Actual	Controller accepts OPEN directly during CLOSING and CLOSE directly during OPENING. Missing safe stop → safety pause → movement in the opposite direction.
Severity	Major
Reproducibility	5/5
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi

	Simulator, Arduino Uno
Evidence / Notes	Video: https://drive.google.com/file/d/1-T6GqTWuxJb73uWLQd_Dz4_pWUYiLqmK/view?usp=sharing

Session Summary

Total tests	**24**	**Passed**	**15**
Failed	**8**	Blocked	1
Bugs reported	**6**	Critical/High	4
Coverage notes		Build result	NO-GO